

A complex network diagram with numerous nodes of varying sizes (black, blue, and grey) connected by thin grey lines. Some nodes are highlighted with larger concentric circles. The background is white with faint grey circular patterns.

Fakultas Ilmu Komputer

Learning is the process of acquiring new understanding, knowledge, behaviors, skills, values, attitudes, and preferences.

Hide Methods *from* Public Access

(Handle HTTP Status Code 403 & 404 from server)



YEREMIA 17:7

Diberkatilah orang yang mengandalkan
TUHAN, yang menaruh harapannya pada
TUHAN!

Harapan adalah bentuk dasar dari kepercayaan akan sesuatu yang diinginkan akan didapatkan atau suatu kejadian akan berbuah kebaikan di waktu yang akan datang.

HARAPAN adalah apa yang Tuhan inginkan terjadi sedangkan **IMPIAN** adalah apa yang kita inginkan terjadi.

Bagaimana caranya hidup mengandalkan Tuhan?

Pertama, Jangan pernah menggantikan

- **SUMBER DAYA** *rohani* (yang tak terbatas) dengan
- sumber daya *duniawi* (yang terbatas) (baca: **Ef. 6:13-18**).

Kedua, **BERUSAHA**lah semaksimal mungkin, tetapi serahkanlah hasilnya pada Tuhan. Mengandalkan Tuhan bukan berarti meniadakan usaha, namun berusaha dengan cara pandang bahwa Tuhanlah yang berdaulat (**Ayb. 42:2**).

Ketiga, **TAAT**lah pada pimpinan Tuhan. Mintalah kekuatan pada Kristus, yang terlebih dulu “*taat sampai mati di kayu salib*” demi menebus dosa kita (**Flp. 2:8**). **Amin**.



**Welcome *to* the Back-end Web
Development *course***

Underscore Notation in Python

```
class Employee:
```

```
    def __init__(self, name, salary):
```

```
        self.name = name
```

Public Member (accessible within or outside of a class)

```
        self._project = project
```

Protected Member (accessible within the class and its sub-classes)

```
        self.__salary = salary
```

Private Member (accessible only within a class)

↑
Data Hiding using Encapsulation

In Python, the use of underscore in variables can have several meanings depending on the context. Conventions (*common practices*) can relate to various aspects, such as **variable naming**, **code structure**, **comment usage**, and so on. Here are some common uses of underscore in Python:

(1) Menandakan variabel "*private*":

In Python, there is no concept of "*private*" variables like in some other programming languages. However, a common convention is to mark a variable as "*private*" by adding an underscore as a prefix to its name. This tells the user that the variable should not be accessed or manipulated from outside the class or module.

(2) Menandakan variabel yang tidak digunakan:

In some situations, we may want to mark an unused variable by using an underscore as its name. This is useful to avoid the "unused variable" warning when running the program.

(3) Menandakan nama variabel yang ambigu:

In some cases, we may want to mark ambiguous or unclear variable names by using underscore.

Underscore Notation *in* PHP

In PHP, the use of the underscore prefix in variable names is one way to mark variables as "private" or "protected" in a class. This convention is also known as "underscore notation".

- ❖ Variables preceded by an underscore are considered "*private*" because they **indicate that they should not be accessed or manipulated directly by code outside the class.**
- ❖ A convention is a set of rules followed by a group of people in a particular field. These conventions are created to ensure consistency, interoperability, and security within a system or environment. Conventions can relate to various aspects, such as variable naming, code structure, use of comments, and so on.

PHP Underscore Prefix

In PHP system development, using underscore prefixes in variable, function, and method names can help to clarify and facilitate code maintenance in the long run. Here are some reasons why the use of underscore prefixes is essential in PHP system development:

1) Menandakan sifat variabel, fungsi, dan metode:

Penggunaan underscore prefix pada nama variabel, fungsi, dan metode dapat membantu menandakan sifat atau hak akses dari item tersebut. Misalnya, variabel yang diawali dengan underscore (seperti `$_private_var`) menunjukkan bahwa variabel tersebut bersifat private dan seharusnya tidak diakses atau dimanipulasi langsung oleh kode di luar kelas atau objek.

2) Mempermudah pembacaan kode:

Penggunaan underscore prefix pada nama variabel, fungsi, dan metode dapat memperjelas bagaimana item tersebut seharusnya digunakan dalam kode. Hal ini dapat memudahkan pembacaan kode oleh pengembang lain atau oleh pengembang yang akan memelihara kode tersebut di masa depan.

3) Memastikan konsistensi dalam kode:

Dengan menggunakan konvensi underscore prefix pada nama variabel, fungsi, dan metode, pengembang dapat memastikan bahwa kode yang dihasilkan konsisten dan mudah dipahami oleh seluruh tim pengembang. Hal ini dapat membantu meningkatkan kualitas kode dan meminimalkan kesalahan dalam pengembangan sistem.

4) Sesuai dengan standar industri:

Penggunaan underscore prefix pada nama variabel, fungsi, dan metode merupakan praktik yang umum dilakukan dalam industri teknologi informasi. Oleh karena itu, pengembang yang menggunakan konvensi ini akan lebih mudah bergabung dengan tim pengembang lain dan beradaptasi dengan standar industri yang berlaku.

Penggunaan Underscore Prefix

Perlu diingat bahwa penggunaan *underscore prefix* pada PHP hanyalah konvensi (*praktik umum*) dan bukan aturan yang harus diikuti. Selain itu, penamaan variabel dan fungsi yang baik harus mencakup lebih dari sekadar penggunaan *underscore prefix*, melainkan juga memperhatikan **keterbacaan**, **kesederhanaan**, dan **konsistensi** dalam kode.

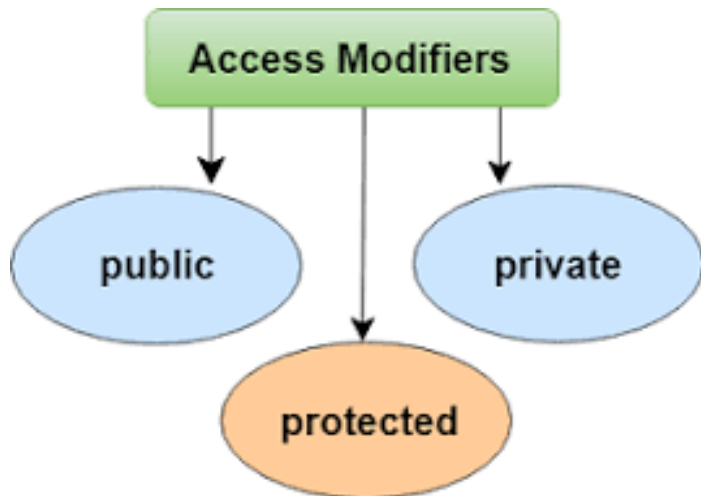
Note:

Penggunaan “*access modifier private*” akan membawa *error* pada *browser*, oleh sebab itu kita butuh penanganan yang tepat supaya tidak terjadi *error*.



Access Modifiers

Access Modifiers	Same Class	Same Package	Sub Class	Other Packages
<i>Public</i>	Y	Y	Y	Y
<i>Protected</i>	Y	Y	Y	N
<i>Private</i>	Y	N	N	N



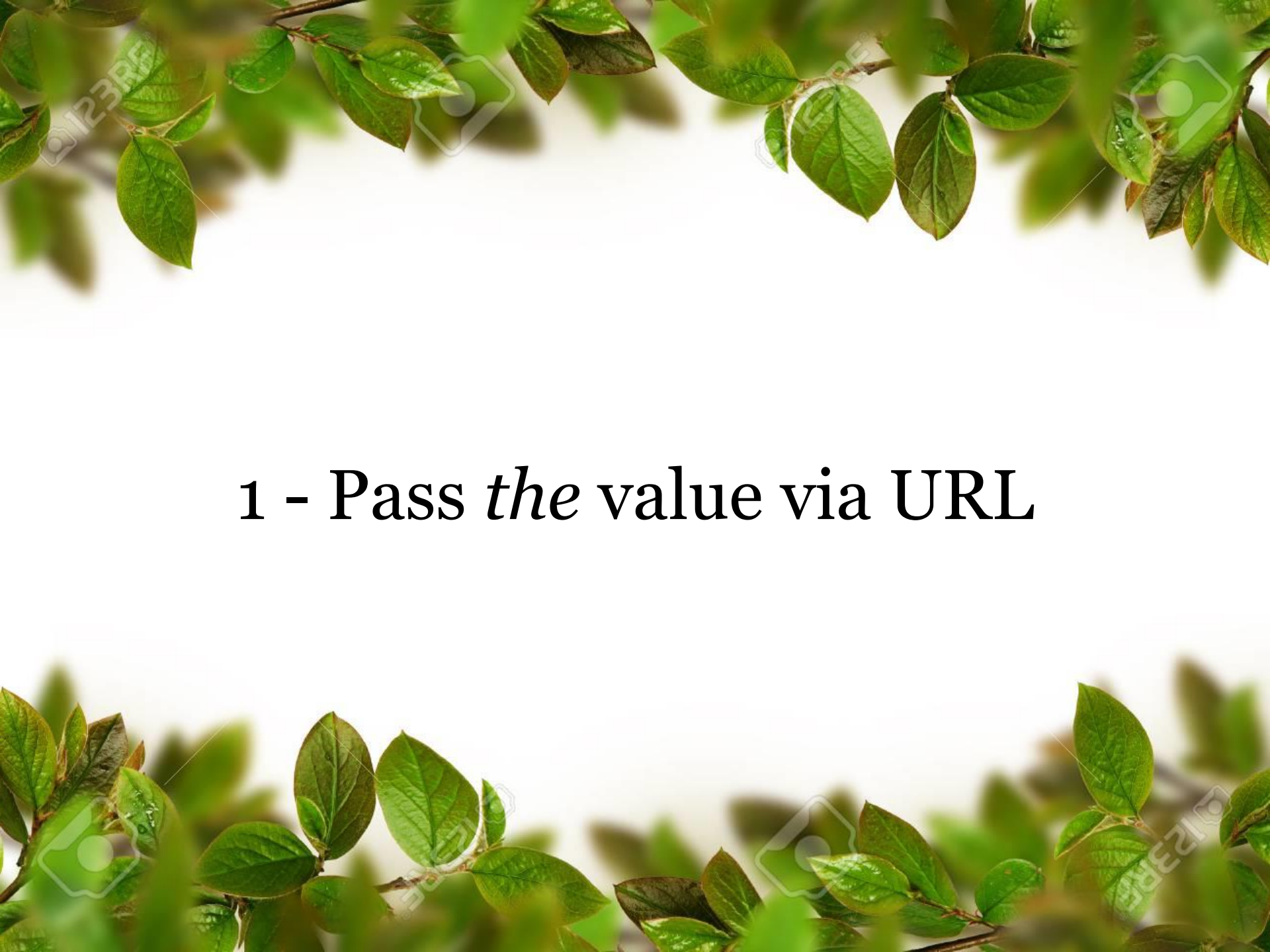
In most of the object-oriented languages access modifiers are used to limit the access to the variables and functions of a class and a package/folder.



Start *from* Here !!!

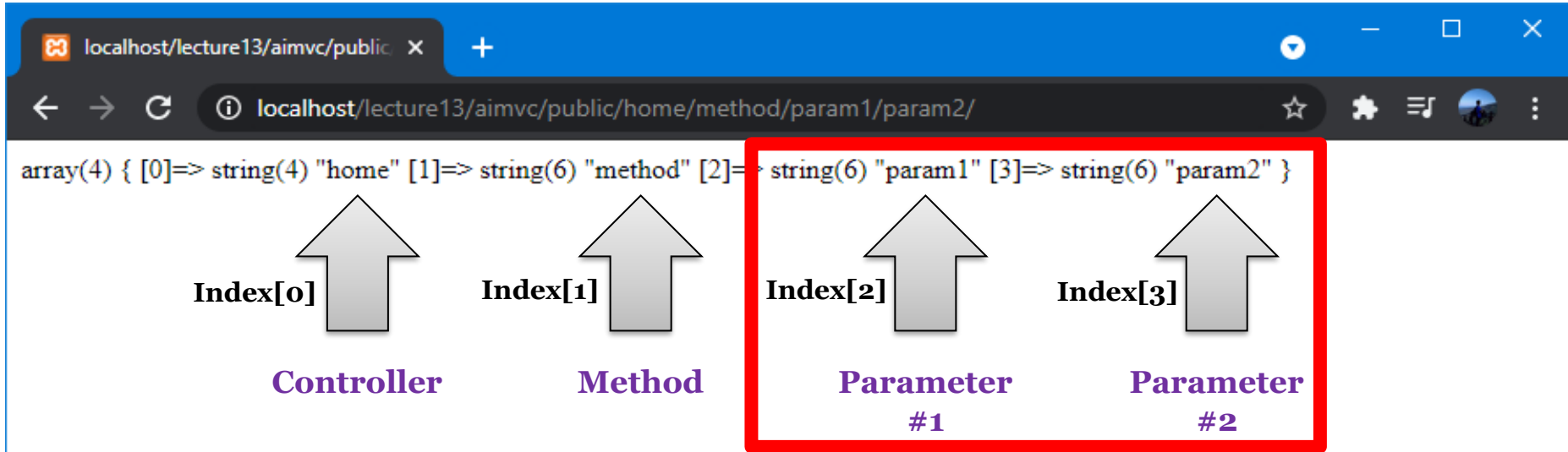


Exercise #1 *for* Students

The slide features a white background with green leaves and branches framing the top and bottom edges. The leaves are vibrant green and appear to be from a tree or shrub. The text is centered in the middle of the slide.

1 - Pass *the* value via URL

Handle Input Data from URL



Our app starts from this directory location:
<http://localhost/lecture13/aimvc/public/>

This means Controller default (*home*) and method default (*index*).

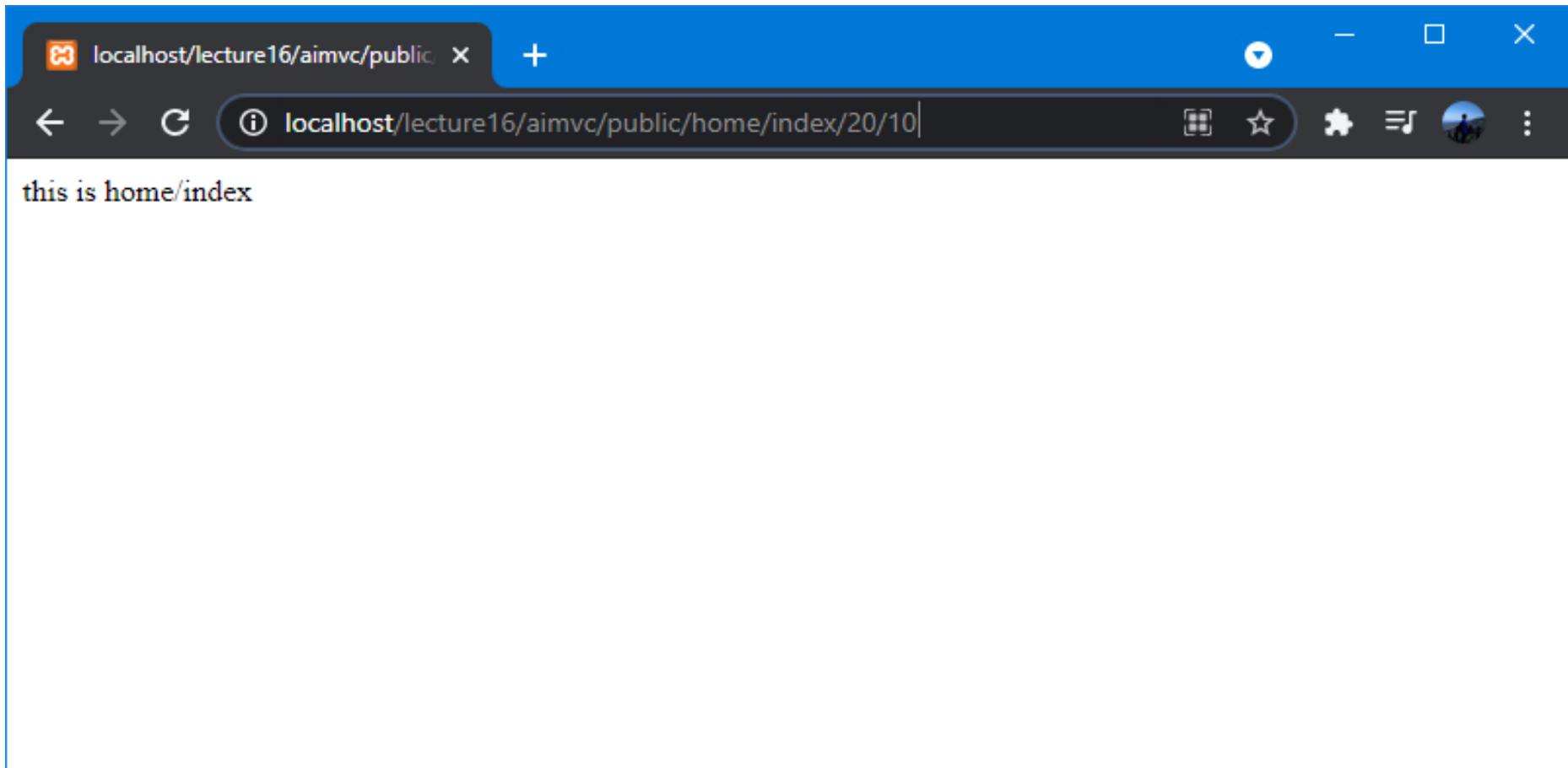
URL Format: <http://localhost/lecture13/aimvc/public/class/method/param1/param2/>

Handle Input Data from Default Method

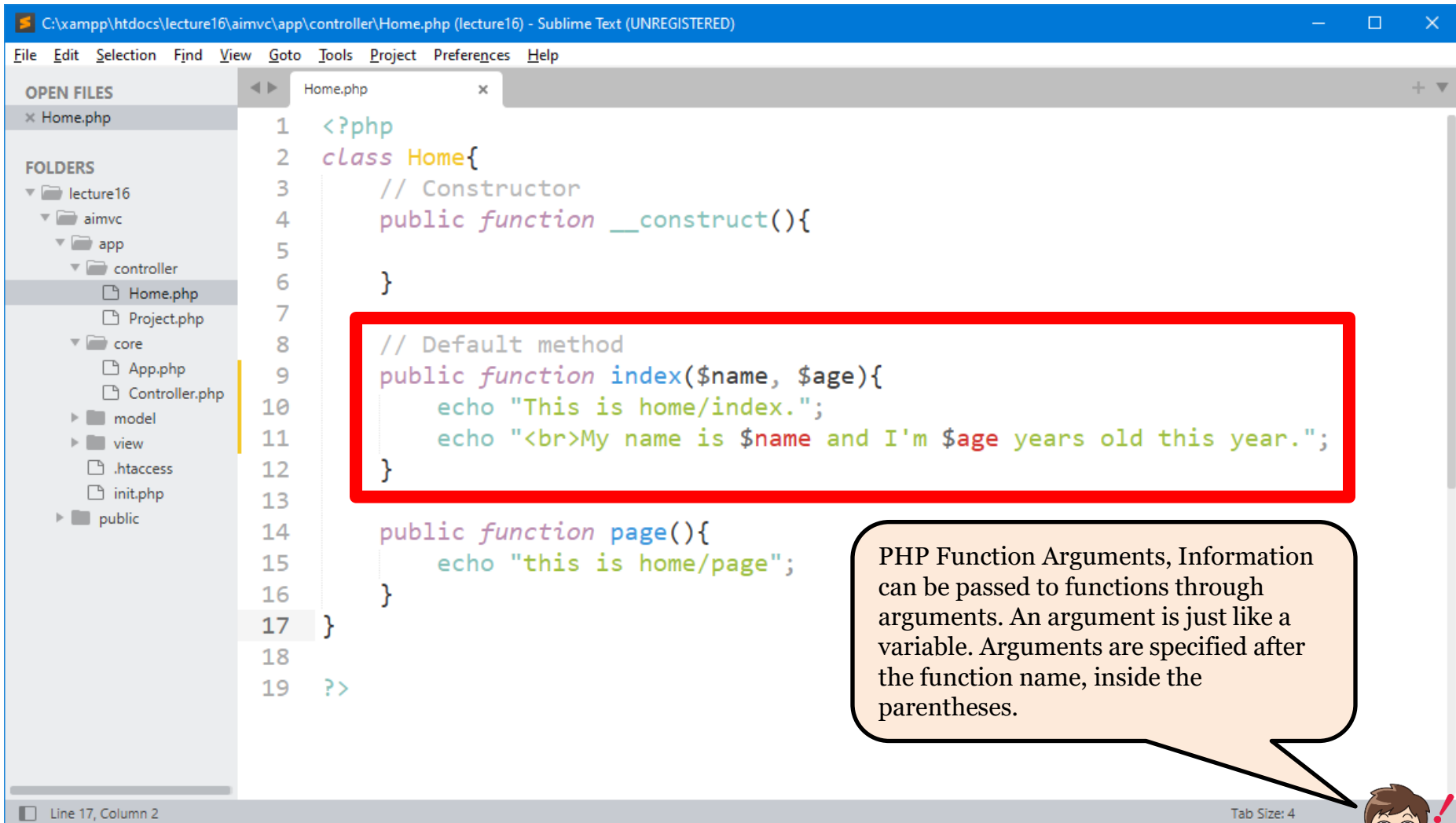
Default method can be accessed through these URLs:

<http://localhost/lecture16/aimvc/public/home/index/>

<http://localhost/lecture16/aimvc/public/home/>



Handle Input Data from Default Method



C:\xampp\htdocs\lecture16\aimvc\app\controller\Home.php (lecture16) - Sublime Text (UNREGISTERED)

File Edit Selection Find View Goto Tools Project Preferences Help

OPEN FILES

- × Home.php

FOLDERS

- lecture16
 - aimvc
 - app
 - controller
 - Home.php
 - Project.php
 - core
 - App.php
 - Controller.php
 - model
 - view
 - .htaccess
 - init.php
 - public

```
1 <?php
2 class Home{
3     // Constructor
4     public function __construct(){
5
6     }
7
8     // Default method
9     public function index($name, $age){
10         echo "This is home/index.";
11         echo "<br>My name is $name and I'm $age years old this year.";
12     }
13
14     public function page(){
15         echo "this is home/page";
16     }
17 }
18
19 ?>
```

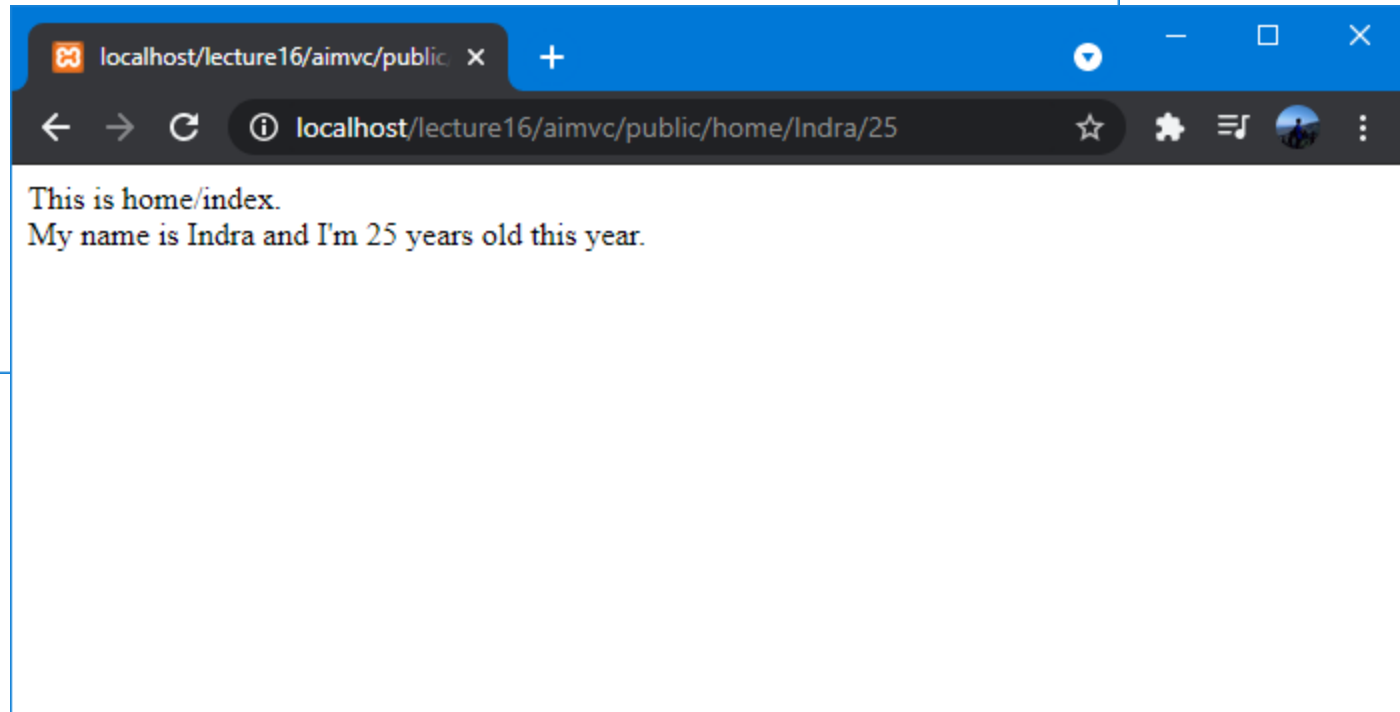
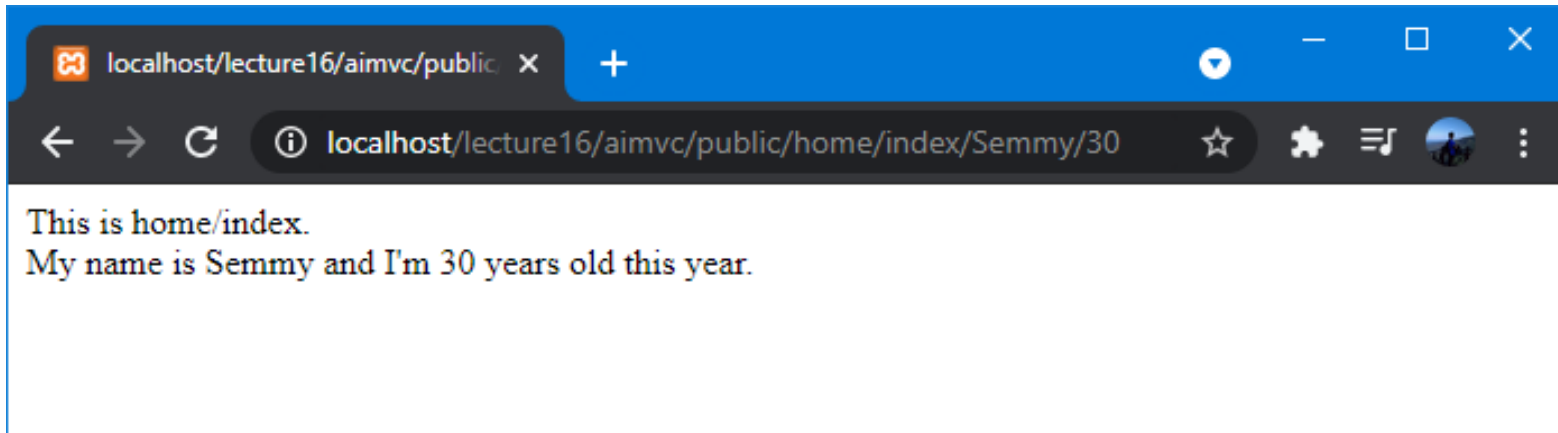
PHP Function Arguments, Information can be passed to functions through arguments. An argument is just like a variable. Arguments are specified after the function name, inside the parentheses.

Line 17, Column 2

Tab Size: 4

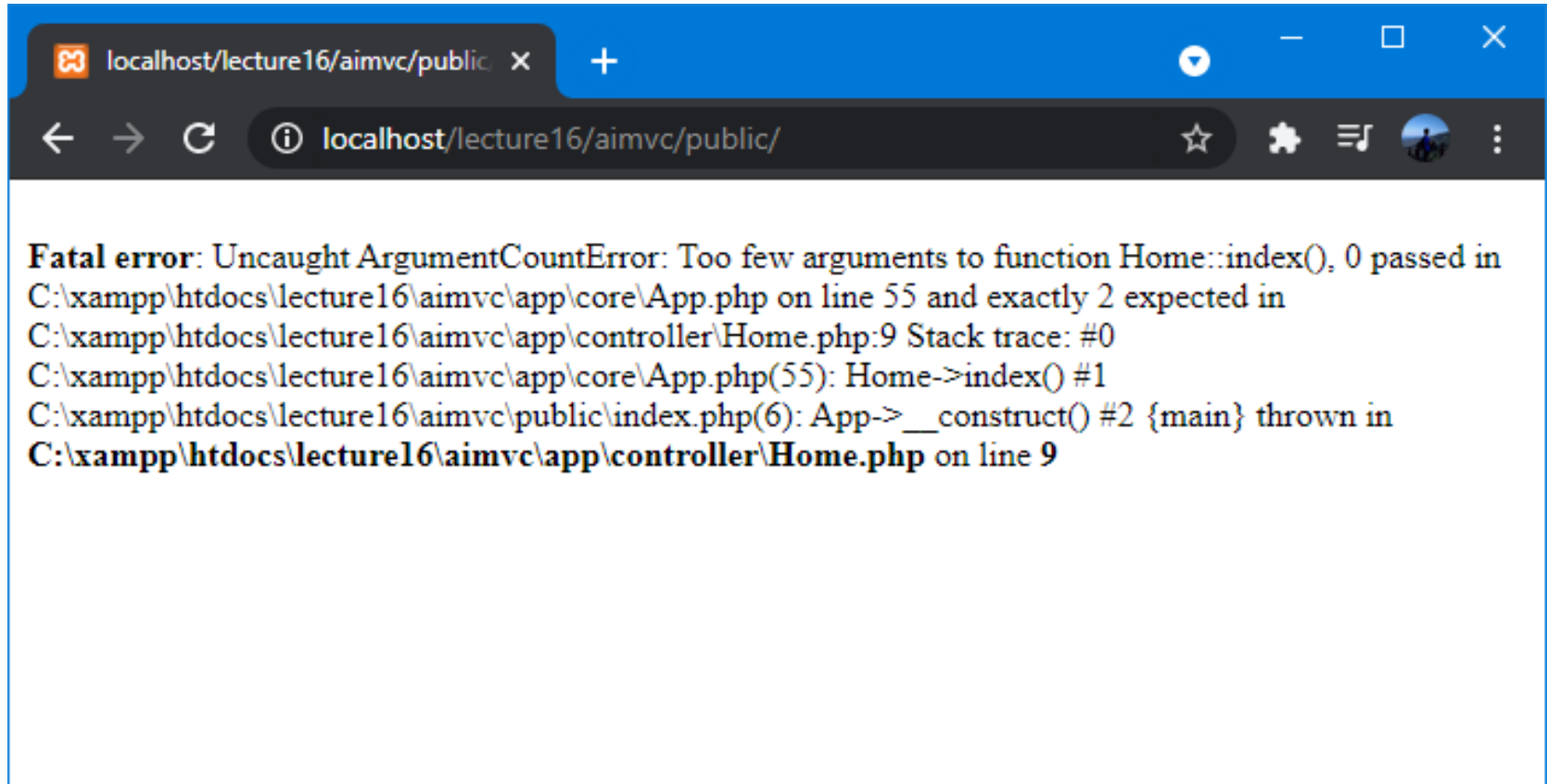


Handle Input Data from Default Method



Why Fatal Error?

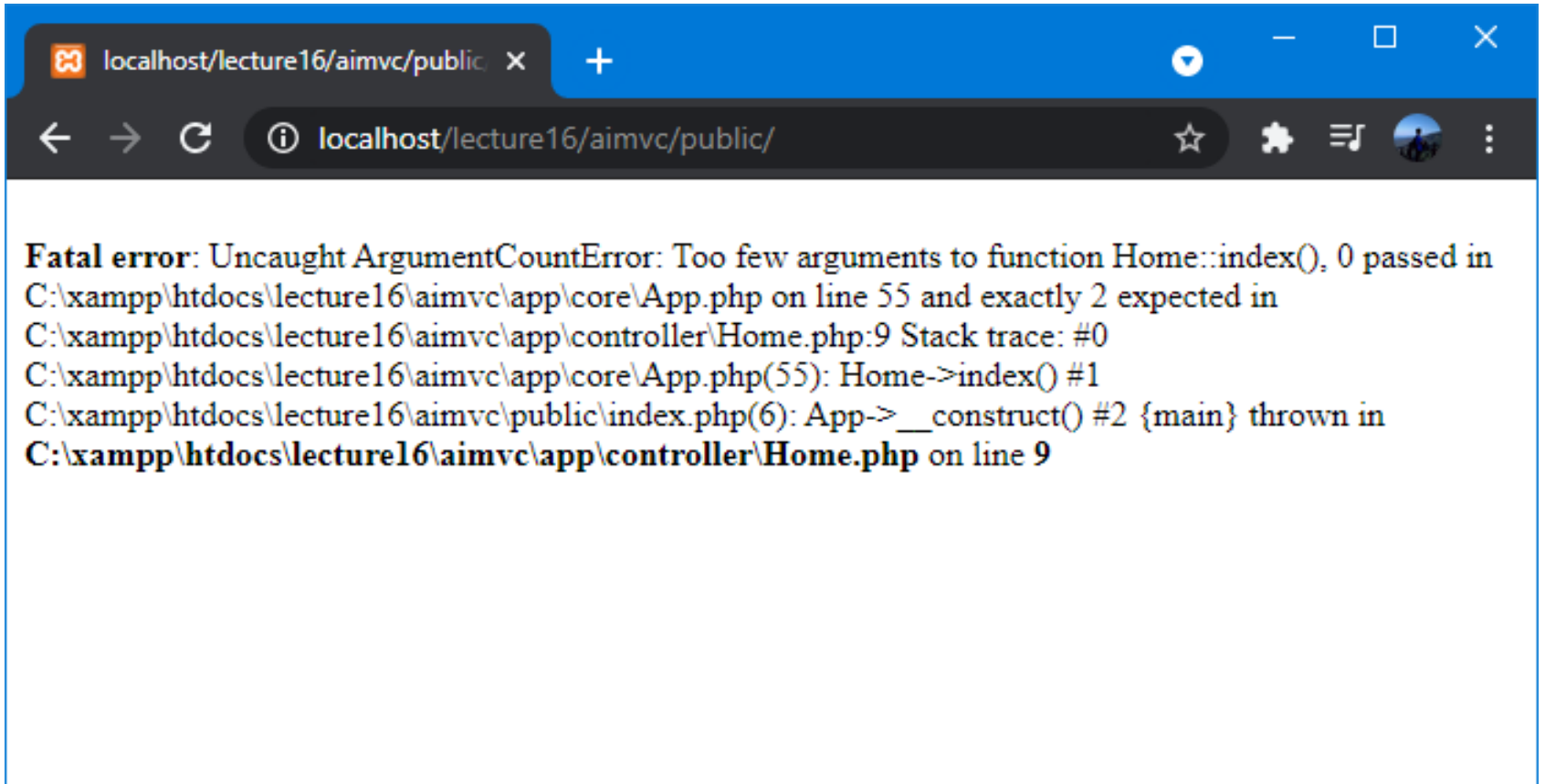
Link URL: *http://localhost/lecture16/aimvc/public/home/*

A screenshot of a web browser window. The address bar shows 'localhost/lecture16/aimvc/public/'. The page content displays a fatal PHP error message. The error message is: 'Fatal error: Uncaught ArgumentCountError: Too few arguments to function Home::index(), 0 passed in C:\xampp\htdocs\lecture16\aimvc\app\core\App.php on line 55 and exactly 2 expected in C:\xampp\htdocs\lecture16\aimvc\app\controller\Home.php:9 Stack trace: #0 C:\xampp\htdocs\lecture16\aimvc\app\core\App.php(55): Home->index() #1 C:\xampp\htdocs\lecture16\aimvc\public\index.php(6): App->__construct() #2 {main} thrown in C:\xampp\htdocs\lecture16\aimvc\app\controller\Home.php on line 9'.

```
Fatal error: Uncaught ArgumentCountError: Too few arguments to function Home::index(), 0 passed in  
C:\xampp\htdocs\lecture16\aimvc\app\core\App.php on line 55 and exactly 2 expected in  
C:\xampp\htdocs\lecture16\aimvc\app\controller\Home.php:9 Stack trace: #0  
C:\xampp\htdocs\lecture16\aimvc\app\core\App.php(55): Home->index() #1  
C:\xampp\htdocs\lecture16\aimvc\public\index.php(6): App->__construct() #2 {main} thrown in  
C:\xampp\htdocs\lecture16\aimvc\app\controller\Home.php on line 9
```

Apakah ada yang tahu mengapa terjadi error?

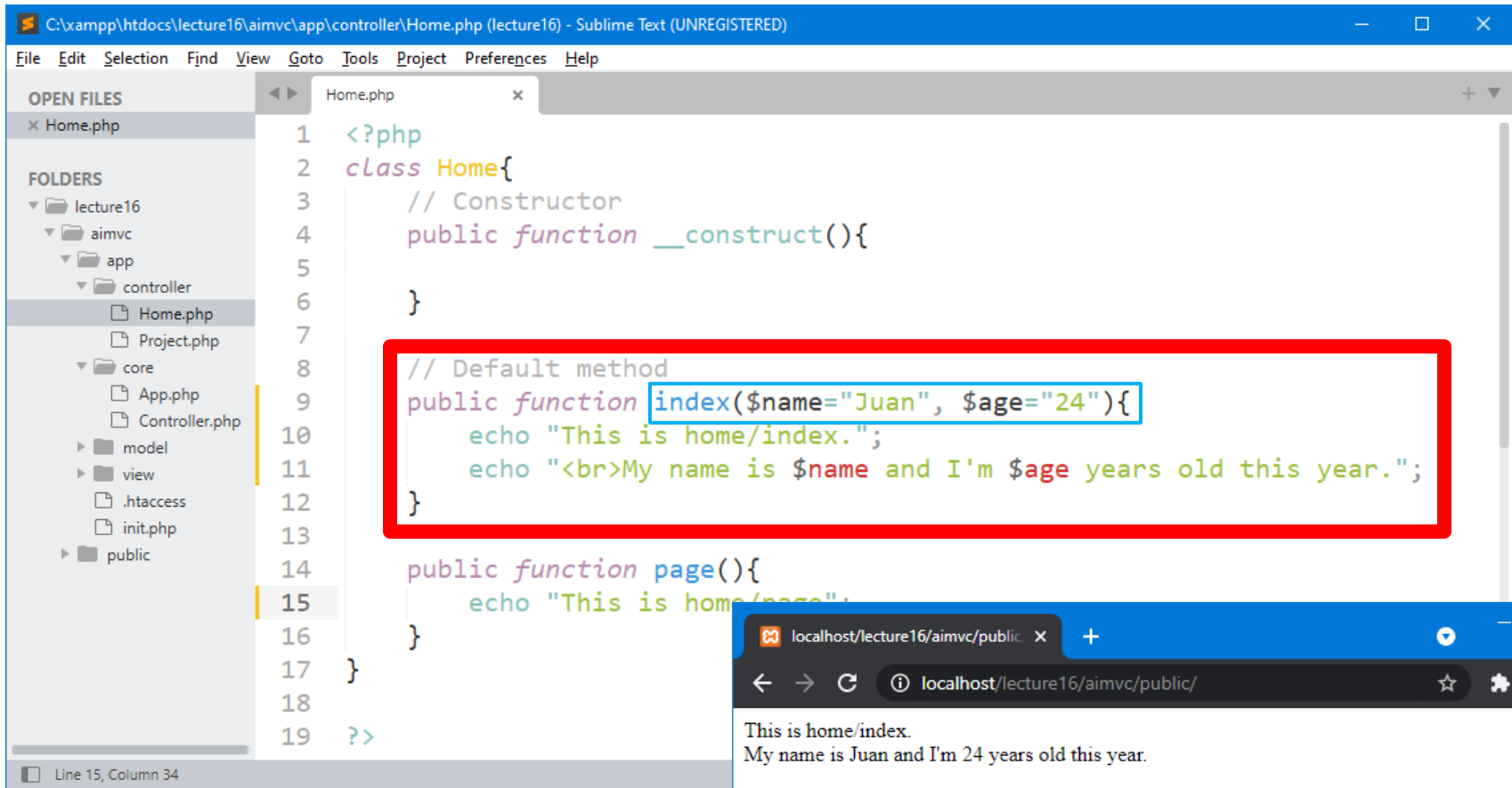
Why Fatal Error?

A screenshot of a web browser window. The address bar shows 'localhost/lecture16/aimvc/public/'. The main content area displays a fatal error message in black text on a white background. The error message reads: 'Fatal error: Uncaught ArgumentCountError: Too few arguments to function Home::index(), 0 passed in C:\xampp\htdocs\lecture16\aimvc\app\core\App.php on line 55 and exactly 2 expected in C:\xampp\htdocs\lecture16\aimvc\app\controller\Home.php:9 Stack trace: #0 C:\xampp\htdocs\lecture16\aimvc\app\core\App.php(55): Home->index() #1 C:\xampp\htdocs\lecture16\aimvc\public\index.php(6): App->__construct() #2 {main} thrown in C:\xampp\htdocs\lecture16\aimvc\app\controller\Home.php on line 9'. The browser's interface includes a blue header bar with a tab labeled 'localhost/lecture16/aimvc/public', navigation buttons (back, forward, refresh), and a dark address bar with the URL 'localhost/lecture16/aimvc/public/'.

```
Fatal error: Uncaught ArgumentCountError: Too few arguments to function Home::index(), 0 passed in  
C:\xampp\htdocs\lecture16\aimvc\app\core\App.php on line 55 and exactly 2 expected in  
C:\xampp\htdocs\lecture16\aimvc\app\controller\Home.php:9 Stack trace: #0  
C:\xampp\htdocs\lecture16\aimvc\app\core\App.php(55): Home->index() #1  
C:\xampp\htdocs\lecture16\aimvc\public\index.php(6): App->__construct() #2 {main} thrown in  
C:\xampp\htdocs\lecture16\aimvc\app\controller\Home.php on line 9
```

There're no default parameters in this method. We may need to remember this, when we create a method with parameters, we should provide the method with “default parameters”.

Default Values for Function Parameters



The image shows a code editor window titled "C:\xampp\htdocs\lecture16\aimvc\app\controller\Home.php (lecture16) - Sublime Text (UNREGISTERED)". The editor displays the following PHP code:

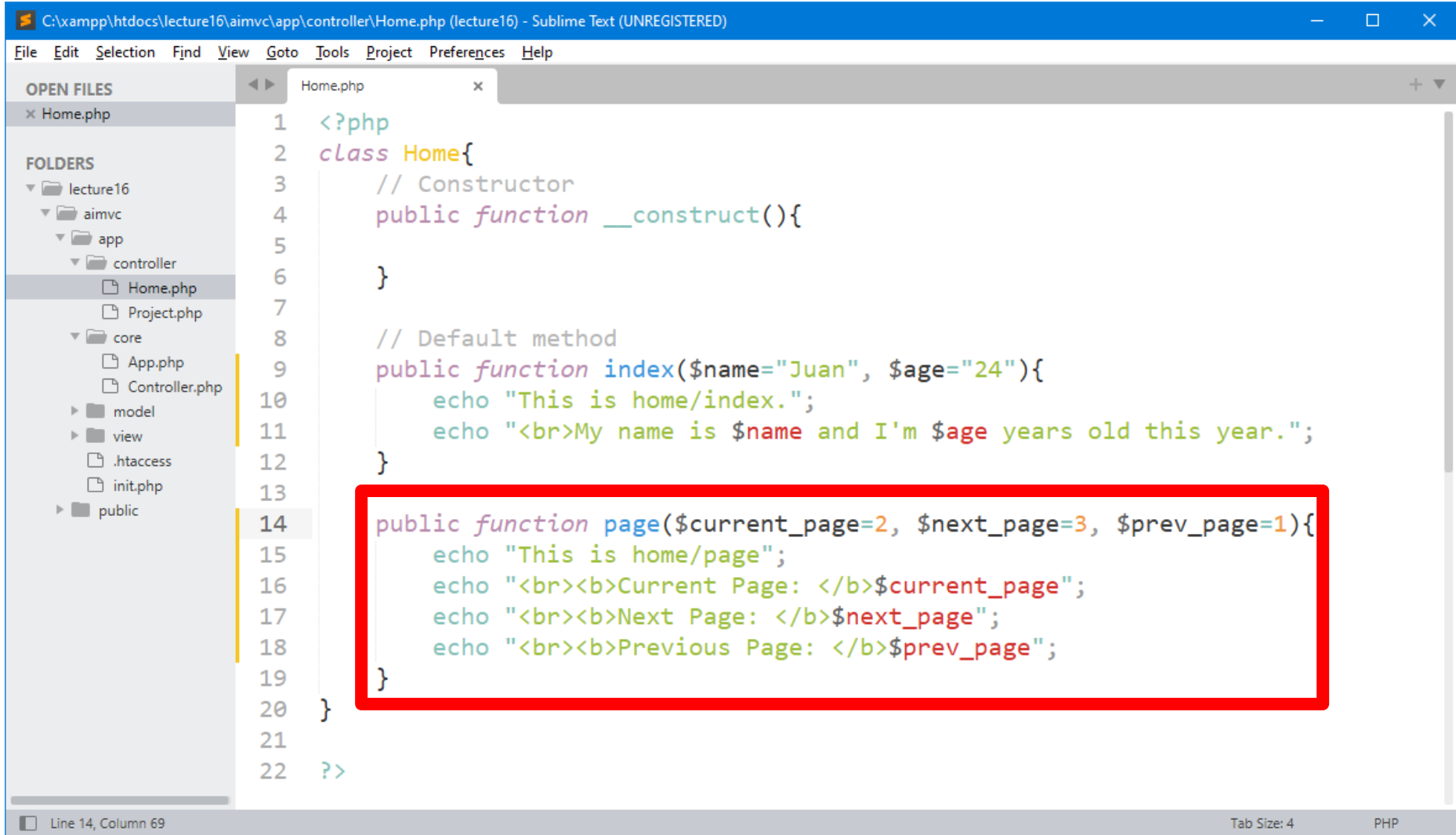
```
1 <?php
2 class Home{
3     // Constructor
4     public function __construct(){
5
6     }
7
8     // Default method
9     public function index($name="Juan", $age="24"){
10         echo "This is home/index.";
11         echo "<br>My name is $name and I'm $age years old this year.";
12     }
13
14     public function page(){
15         echo "This is home/page.";
16     }
17 }
18
19 ?>
```

The code is highlighted with a red box, and the function signature `public function index($name="Juan", $age="24")` is highlighted with a blue box. The browser window shows the output of the `index` method:

localhost/lecture16/aimvc/public/

This is home/index.
My name is Juan and I'm 24 years old this year.

Many Arguments from URL Request



```
C:\xampp\htdocs\lecture16\aimvc\app\controller\Home.php (lecture16) - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help

OPEN FILES
x Home.php

FOLDERS
lecture16
  aimvc
    app
      controller
        Home.php
        Project.php
      core
        App.php
        Controller.php
      model
      view
      .htaccess
      init.php
      public

1  <?php
2  class Home{
3      // Constructor
4      public function __construct(){
5
6      }
7
8      // Default method
9      public function index($name="Juan", $age="24"){
10         echo "This is home/index.";
11         echo "<br>My name is $name and I'm $age years old this year.";
12     }
13
14     public function page($current_page=2, $next_page=3, $prev_page=1){
15         echo "This is home/page";
16         echo "<br><b>Current Page: </b>$current_page";
17         echo "<br><b>Next Page: </b>$next_page";
18         echo "<br><b>Previous Page: </b>$prev_page";
19     }
20 }
21
22 ?>
```

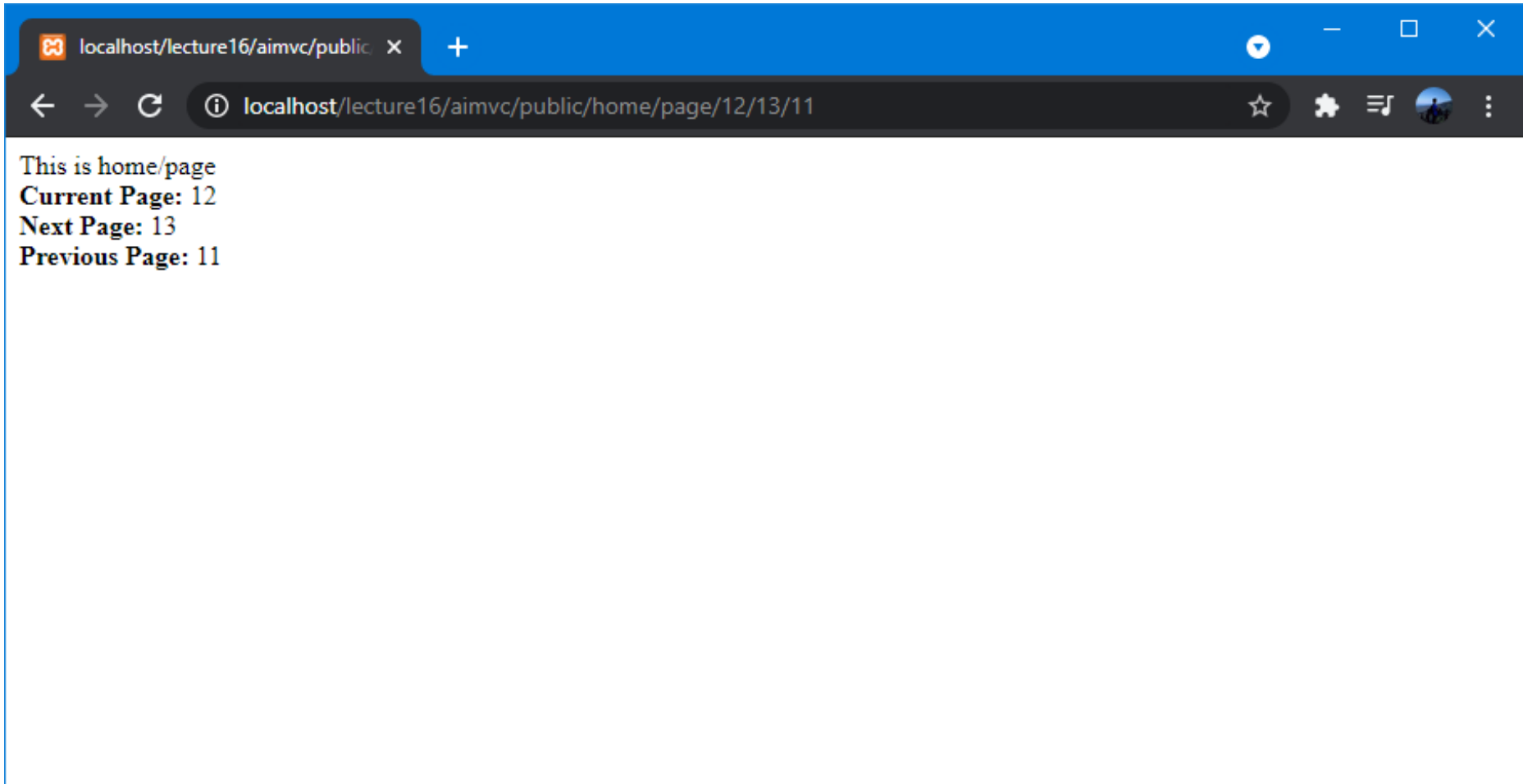
Line 14, Column 69

Tab Size: 4

PHP

You can add as many arguments as you want, just separate them with a comma.

Many Arguments from URL Request

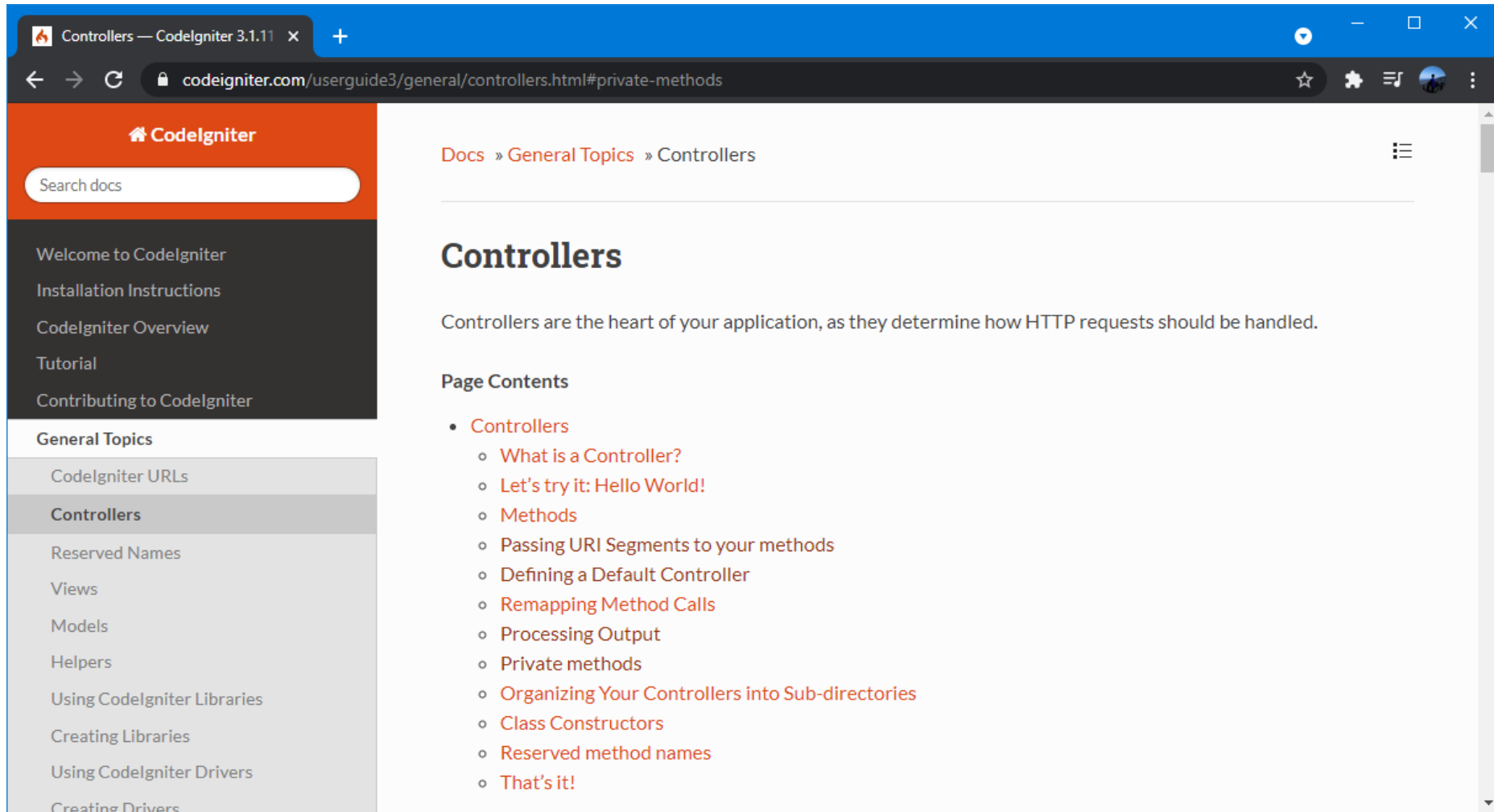


We added three arguments as a parameter input for the method in class controller, we separate data with a comma.

The slide features a white background with green leaves and branches framing the top and bottom edges. The leaves are vibrant green with some brownish edges, suggesting they might be from a tree like a Japanese Quince.

2 - Prevent Direct Access Methods

Controllers *in* CodeIgniter



The screenshot shows a web browser window displaying the CodeIgniter 3.1.11 documentation page for Controllers. The browser's address bar shows the URL `codeigniter.com/userguide3/general/controllers.html#private-methods`. The page has a blue header with the CodeIgniter logo and a search bar. The left sidebar contains a navigation menu with links to 'Welcome to CodeIgniter', 'Installation Instructions', 'CodeIgniter Overview', 'Tutorial', 'Contributing to CodeIgniter', and 'General Topics'. The 'General Topics' section is expanded, showing a list of topics including 'CodeIgniter URLs', 'Controllers', 'Reserved Names', 'Views', 'Models', 'Helpers', 'Using CodeIgniter Libraries', 'Creating Libraries', 'Using CodeIgniter Drivers', and 'Creating Drivers'. The 'Controllers' topic is selected and highlighted. The main content area shows the breadcrumb 'Docs » General Topics » Controllers' and the title 'Controllers'. Below the title, a paragraph states: 'Controllers are the heart of your application, as they determine how HTTP requests should be handled.' A 'Page Contents' section follows, listing the following topics:

- Controllers
 - What is a Controller?
 - Let's try it: Hello World!
 - Methods
 - Passing URI Segments to your methods
 - Defining a Default Controller
 - Remapping Method Calls
 - Processing Output
 - Private methods
 - Organizing Your Controllers into Sub-directories
 - Class Constructors
 - Reserved method names
 - That's it!

Link: <https://codeigniter.com/userguide3/general/controllers.html>

**How to prevent direct
access methods in *controller*
class from URL?**



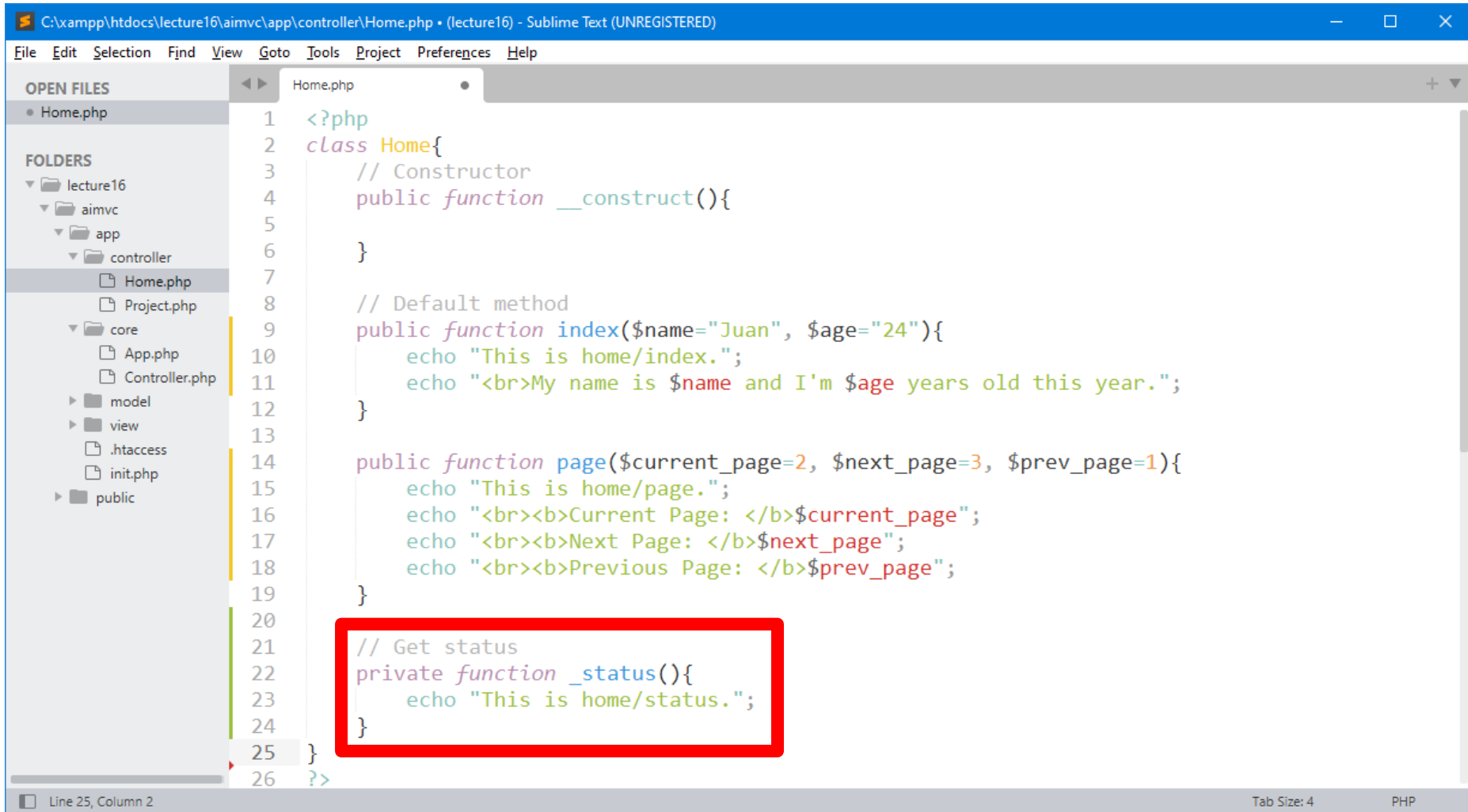
How to Prevent Direct Access Methods?

In some cases, we may want certain methods **hidden from public access**, so there's no direct access to a method. In order to achieve this, simply declare the method as being ***private*** or ***protected*** and it will not be served via a URL request.



We can create *private* or *protected* functions in the class controller by using the ***underscore prefix***.

Close Direct Access - *Private Methods*



```
C:\xampp\htdocs\lecture16\aimvc\app\controller\Home.php • (lecture16) - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help

OPEN FILES
• Home.php

FOLDERS
▼ lecture16
  ▼ aimvc
    ▼ app
      ▼ controller
        Home.php
        Project.php
      ▼ core
        App.php
        Controller.php
      ▼ model
      ▼ view
        .htaccess
        init.php
      ▼ public

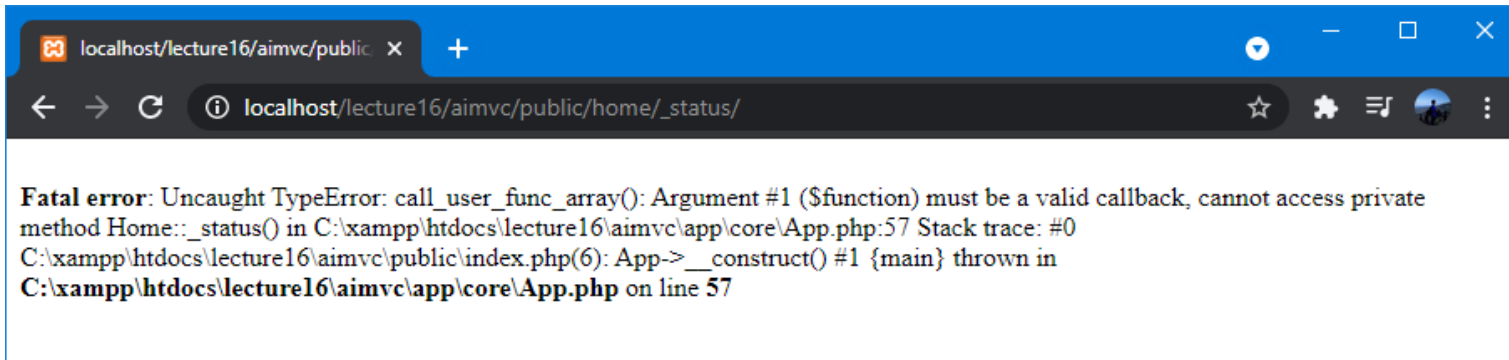
1  <?php
2  class Home{
3      // Constructor
4      public function __construct(){
5
6      }
7
8      // Default method
9      public function index($name="Juan", $age="24"){
10         echo "This is home/index.";
11         echo "<br>My name is $name and I'm $age years old this year.";
12     }
13
14     public function page($current_page=2, $next_page=3, $prev_page=1){
15         echo "This is home/page.";
16         echo "<br><b>Current Page: </b>$current_page";
17         echo "<br><b>Next Page: </b>$next_page";
18         echo "<br><b>Previous Page: </b>$prev_page";
19     }
20
21     // Get status
22     private function _status(){
23         echo "This is home/status.";
24     }
25 }
26 ?>
```

Line 25, Column 2

Tab Size: 4 PHP

Prefixing method names with an ***underscore*** will also prevent them from being called from URL. This is a legacy feature that is left for backwards-compatibility.

Try to access private method via the URL

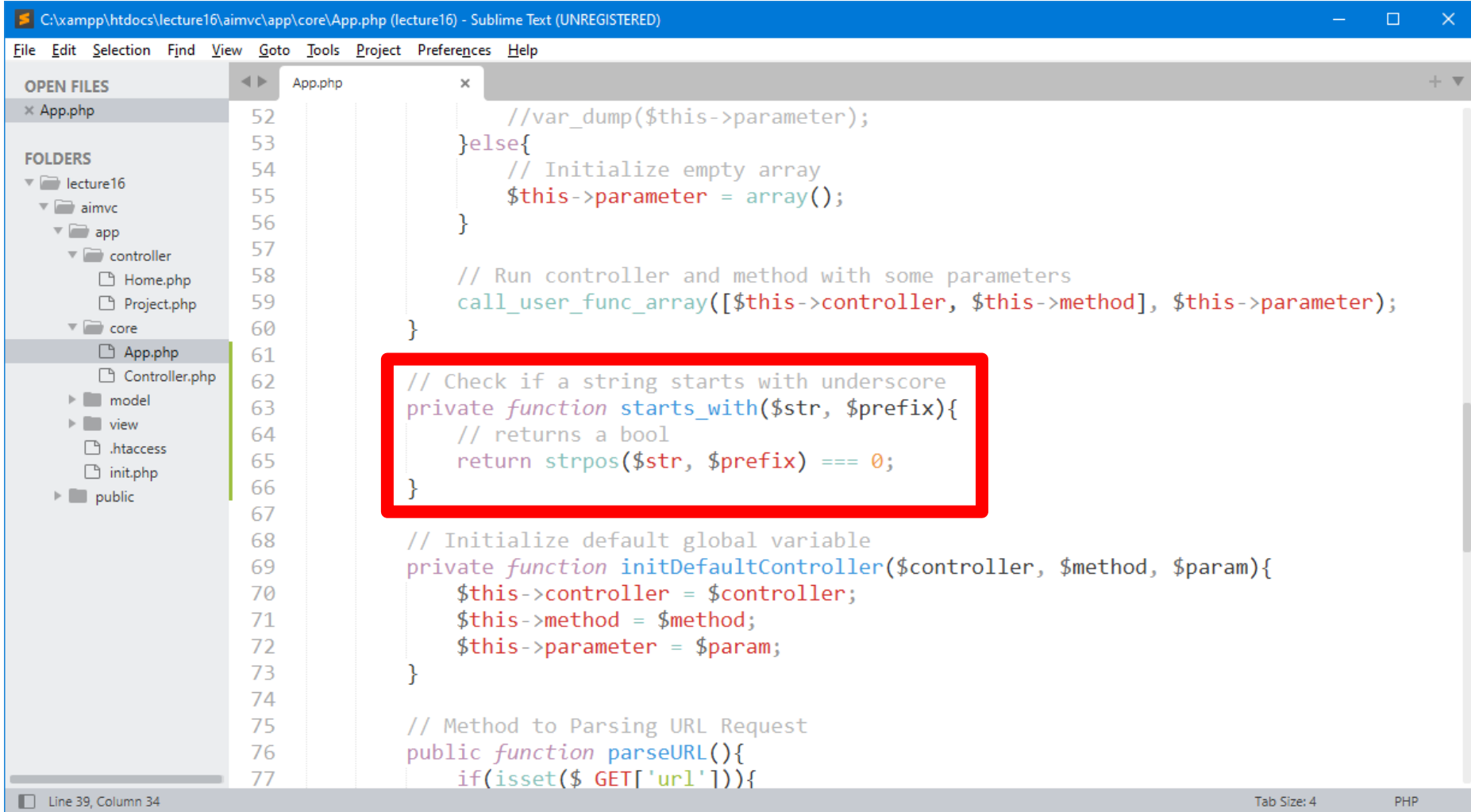


Bagaimana agar supaya tidak muncul error seperti ini, ketika users akses private method? Dan memang kita semua tidak suka jika ada error seperti ini pada saat pengembangan web.

Browser returns an fatal error must be a valid callback, *cannot access private method*.



Create a Function to Check Underscore



C:\xampp\htdocs\lecture16\aimvc\app\core\App.php (lecture16) - Sublime Text (UNREGISTERED)

File Edit Selection Find View Goto Tools Project Preferences Help

OPEN FILES

- x App.php

FOLDERS

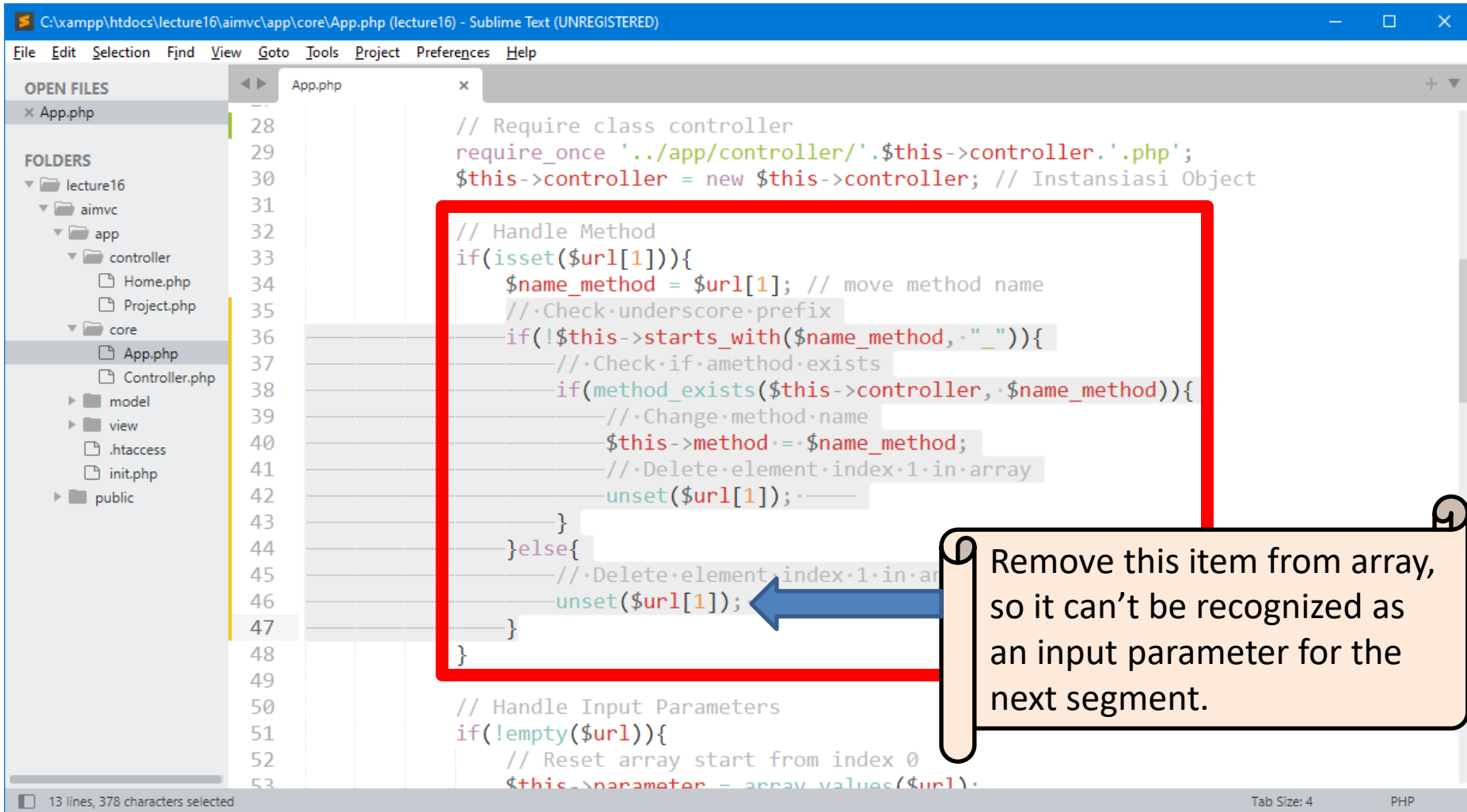
- lecture16
 - aimvc
 - app
 - controller
 - Home.php
 - Project.php
 - core
 - App.php**
 - Controller.php
 - model
 - view
 - .htaccess
 - init.php
 - public

```
52 //var_dump($this->parameter);
53 }else{
54 // Initialize empty array
55 $this->parameter = array();
56 }
57
58 // Run controller and method with some parameters
59 call_user_func_array([$this->controller, $this->method], $this->parameter);
60 }
61
62 // Check if a string starts with underscore
63 private function starts_with($str, $prefix){
64 // returns a bool
65 return strpos($str, $prefix) === 0;
66 }
67
68 // Initialize default global variable
69 private function initDefaultController($controller, $method, $param){
70 $this->controller = $controller;
71 $this->method = $method;
72 $this->parameter = $param;
73 }
74
75 // Method to Parsing URL Request
76 public function parseURL(){
77 if(isset($_GET['url'])){
```

Line 39, Column 34

Tab Size: 4 PHP

Update *your* Core App.php File



C:\xampp\htdocs\lecture16\aimvc\app\core\App.php (lecture16) - Sublime Text (UNREGISTERED)

File Edit Selection Find View Goto Tools Project Preferences Help

OPEN FILES
x App.php

FOLDERS
lecture16
aimvc
app
controller
Home.php
Project.php
core
App.php
Controller.php
model
view
.htaccess
init.php
public

```
28 // Require class controller
29 require_once '../app/controller/'.$this->controller.'.php';
30 $this->controller = new $this->controller; // Instansiasi Object
31
32 // Handle Method
33 if(isset($url[1])){
34     $name_method = $url[1]; // move method name
35     //Check underscore prefix
36     if(!$this->starts_with($name_method, "_")){
37         //Check if a method exists
38         if(method_exists($this->controller, $name_method)){
39             //Change method name
40             $this->method = $name_method;
41             //Delete element index 1 in array
42             unset($url[1]);
43         }
44     }else{
45         //Delete element index 1 in array
46         unset($url[1]);
47     }
48 }
49
50 // Handle Input Parameters
51 if(!empty($url)){
52     // Reset array start from index 0
53     $this->parameter = array_values($url);
```

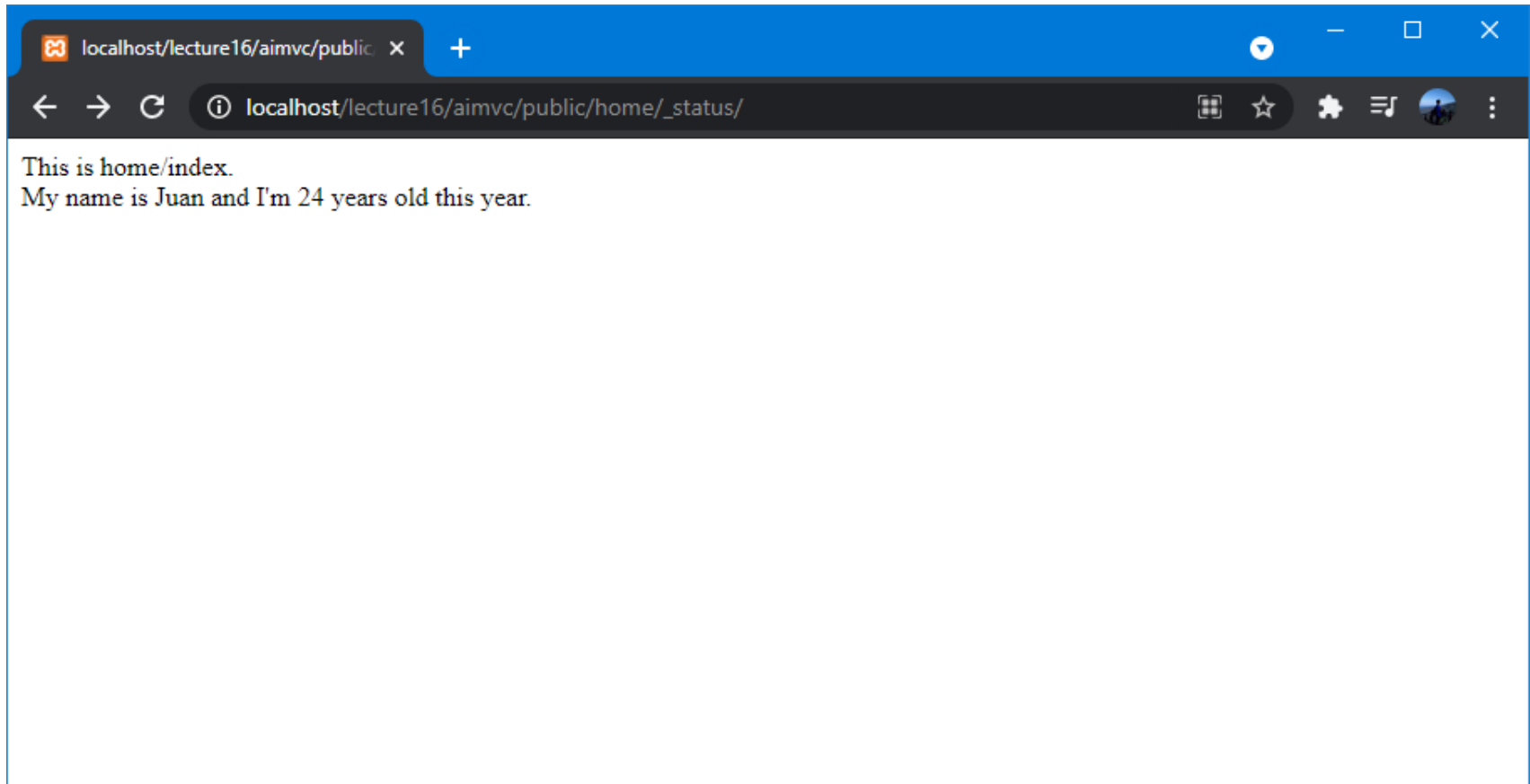
Remove this item from array, so it can't be recognized as an input parameter for the next segment.

13 lines, 378 characters selected

Tab Size: 4 PHP

Expected Output ()

Handle Direct Access to a Private Method



Now, we can completely Prevent Direct Access to a private method from URL and even try to handle underscore prefix from core *App.php* file.

**Lakukan uji coba di
local PC masing-
masing.**

Latihan untuk *Students*

Create directory : C:\xampp\htdocs\lecture20\
Create file : __.php

Buat semua contoh yang berhubungan dengan AIMVC project (sudah di pelajari sebelumnya).

**Upload code ke sisi
server untuk di
Demo.**

Create Directory in Server

[1] Masuk sebagai *administrator* atau *root*:

➤ ***sudo su***

[2] Buat project folder baru dengan perintah (Parent directory):

➤ ***sudo mkdir -p /var/www/html/lecture20***

[3] Konfigurasi permisi hak akses dan pemilik dari folder:

➤ ***sudo chmod -R 755 /var/www/html/lecture20***

➤ ***sudo chown -R semmy:semmy /var/www/html/lecture20***

[4] Restart Apache Web Server:

➤ ***sudo systemctl restart apache2***

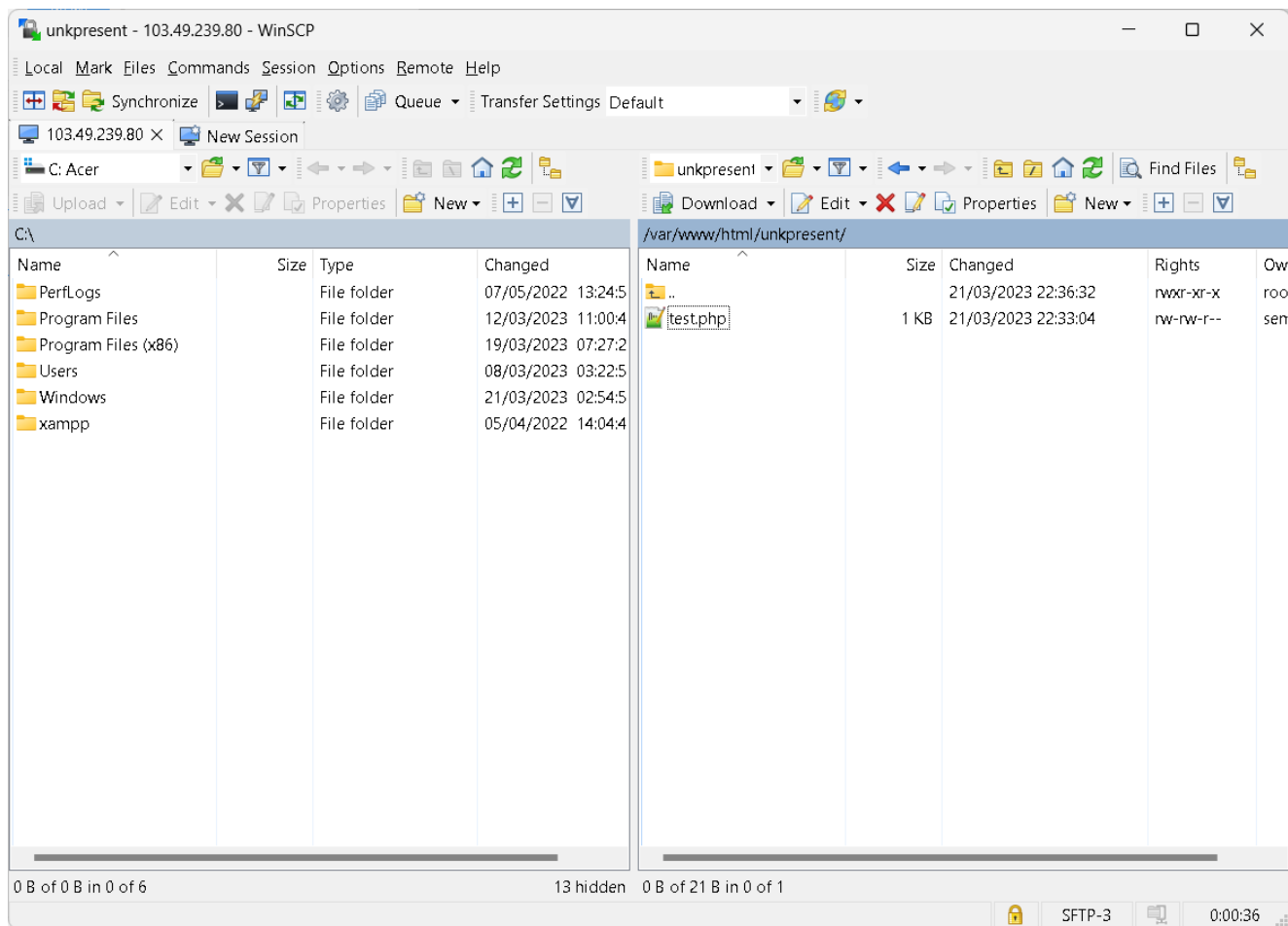
Note:

Ubah nama user “**semmy**” dengan nama *user* access ubuntu kalian.

```
root@unklab: /var/www/html
root@unklab:/home/semmy# sudo mkdir -p /var/www/html/unkpresent
root@unklab:/home/semmy# sudo chmod -R 755 /var/www/html/unkpresent
root@unklab:/home/semmy# cd ..
root@unklab:/home# cd ..
root@unklab:/# ls
bin  dev  home  lib32  libx32  media  opt  root  sbin  srv  tmp  var
boot  etc  lib  lib64  lost+found  mnt  proc  run  snap  sys  usr
root@unklab:/# cd var
root@unklab:/var# ls
backups  crash  local  log  opt  snap  tmp
cache  lib  lock  mail  run  spool  www
root@unklab:/var# cd www/
root@unklab:/var/www# ls
html
root@unklab:/var/www# cd html/
root@unklab:/var/www/html# ls
index.html  test.php  unkpresent
root@unklab:/var/www/html# ls -l
total 20
-rw-r--r-- 1 root root 10671 Mar 26 07:06 index.html
-rw-r--r-- 1 root root 20 Mar 26 07:29 test.php
drwxr-xr-x 2 root root 4096 Mar 26 07:30 unkpresent
root@unklab:/var/www/html# sudo chown -R semmy:semmy /var/www/html/unkpresent
root@unklab:/var/www/html# sudo systemctl restart apache2
root@unklab:/var/www/html# ls -l
total 20
-rw-r--r-- 1 root root 10671 Mar 26 07:06 index.html
-rw-r--r-- 1 root root 20 Mar 26 07:29 test.php
drwxr-xr-x 2 semmy semmy 4096 Mar 26 07:30 unkpresent
root@unklab:/var/www/html#
```

Upload Project via WinSCP

- 1) Buka WinSCP dan arahkan ke “/var/www/html/lecture20”.
- 2) Selanjutnya, upload **file code lecture#20** dari local PC dan ke directory server menggunakan WinSCP.
- 3) Buka via Browser.



Output ditampilkan menggunakan domain name dan bisa di akses secara public via Browser:

Contohnya:

<https://domainkelompok.site/lecture20/aimvc/public/>



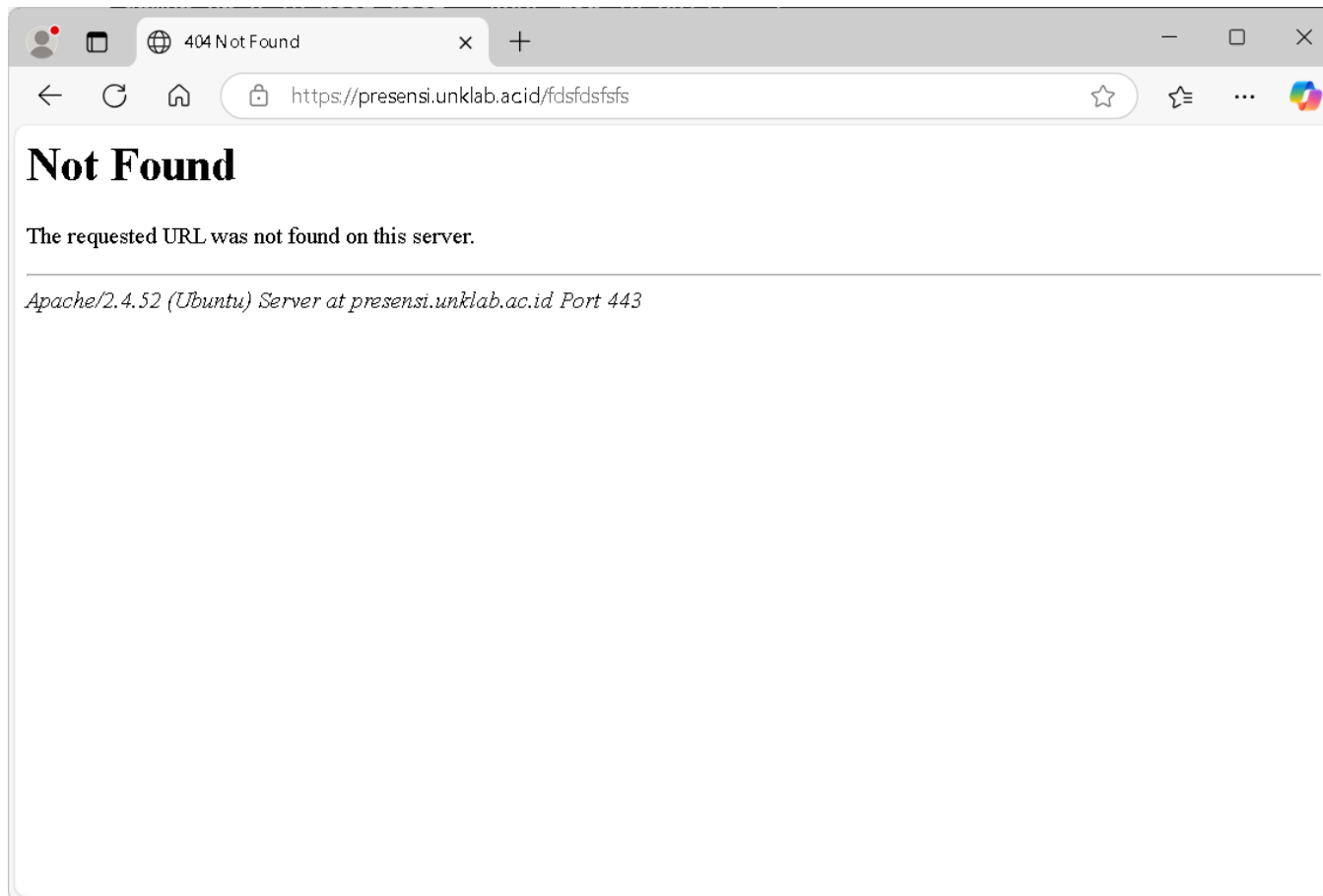
Exercise #2 *for* Students

Handle HTTP error code **404** (Not Found) and **403** (Forbidden) using .htaccess Configuration file.

HTTP Status Code 404 (Not Found)

Coba ketikan secara random sesudah domain name kalian, contohnya:

<https://presensi.unklab.ac.id/fdsfdsfsfs>



HTTP Status Code 403 (Forbidden)

Selanjutnya, Coba access private directory dari URL, apakah bisa di akses??? Mengapa, contohnya:

<https://presensi.unklab.ac.id/unklab/app/>

Forbidden

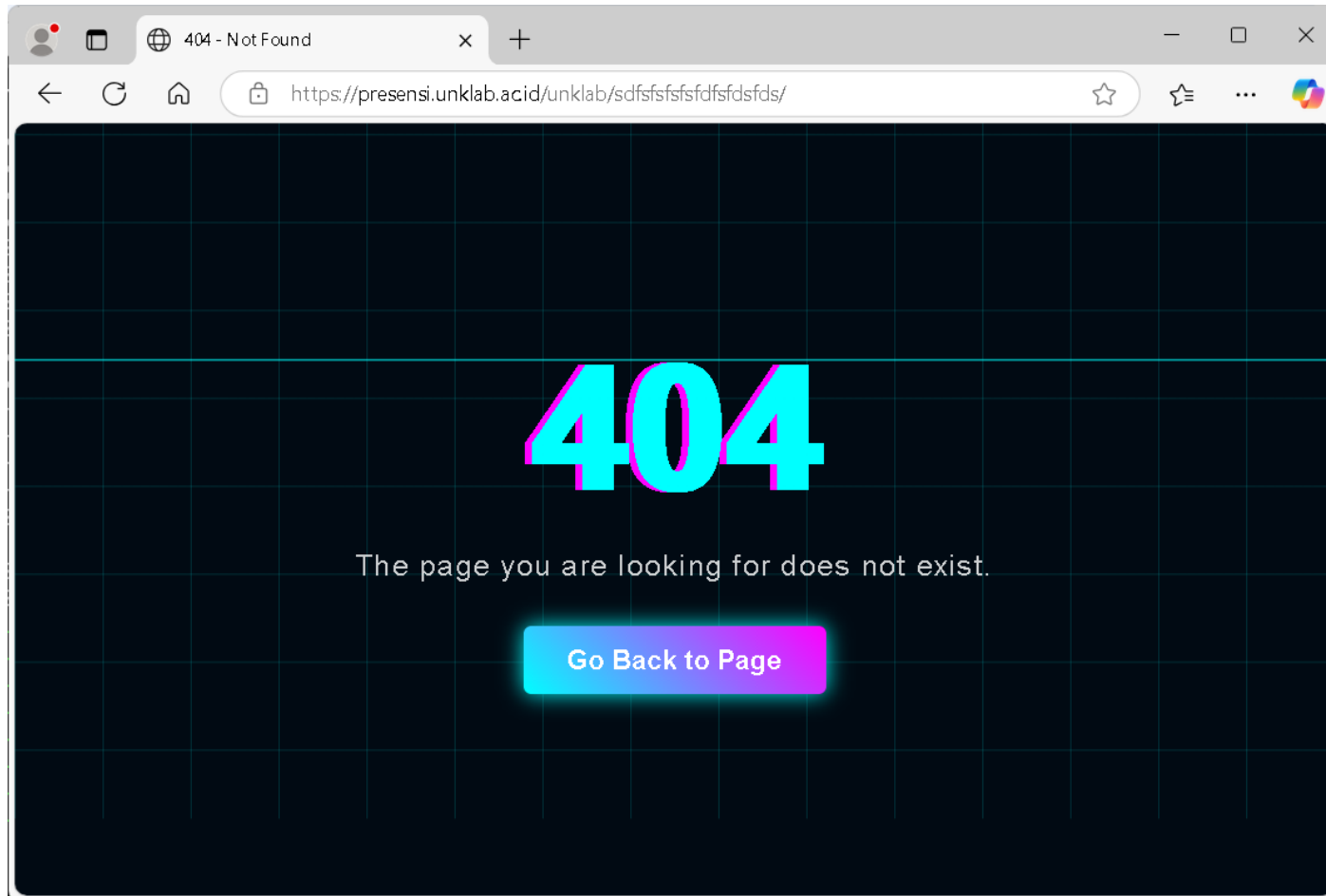
You don't have permission to access this resource.

Apache/2.4.47 (Win64) OpenSSL/1.1.1k PHP/8.0.6 Server at localhost Port 80

HTTP Status Code 404 (Not Found)

Sesudah dihandle, coba ketikan secara random sesudah domain name kalian, contohnya:

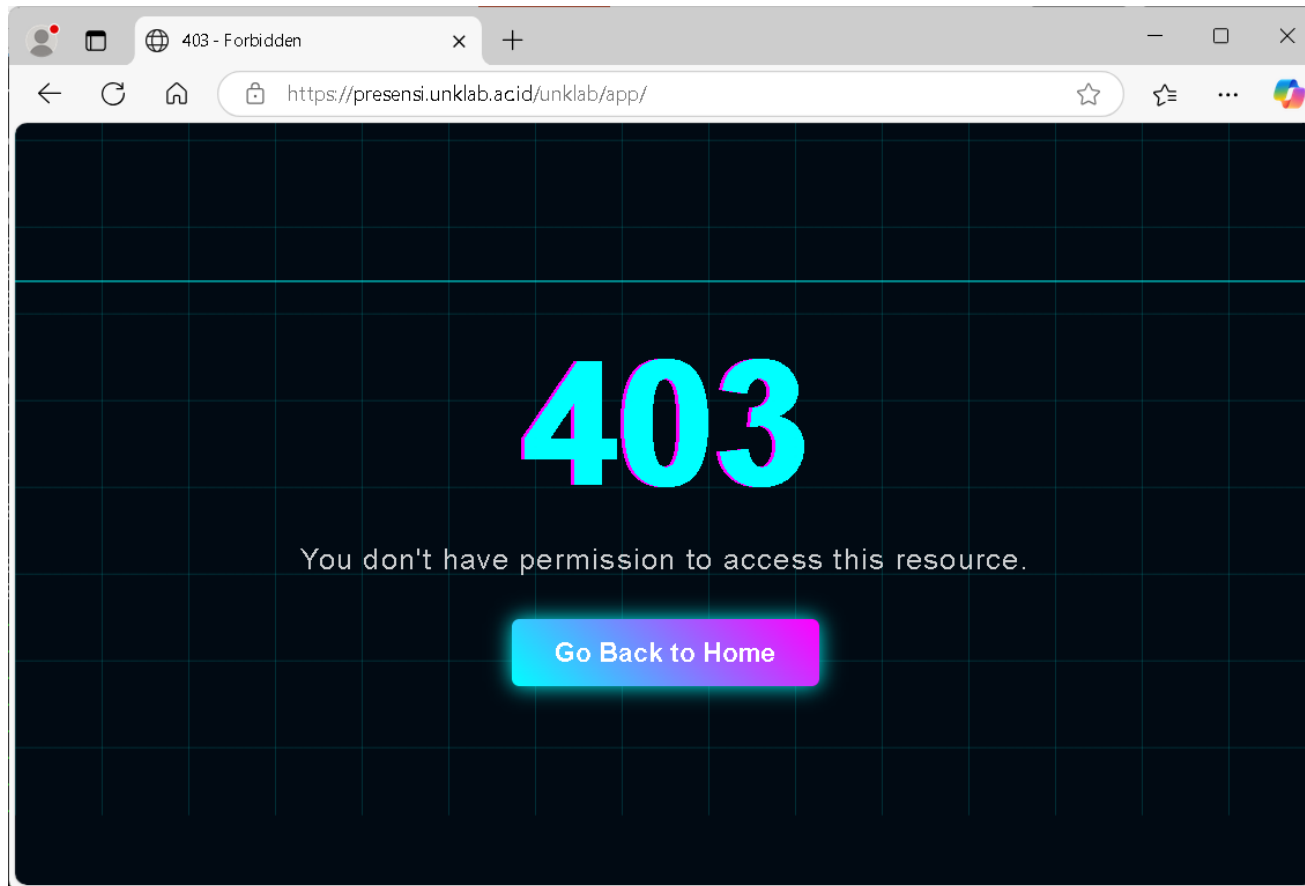
<https://presensi.unklab.ac.id/unklab/sdfsfsfsfsfdfsfdsfds/>



HTTP Status Code 403 (Forbidden)

Sesudah dihandle, coba access kembali private directory dari URL,

<https://presensi.unklab.ac.id/unklab/app/>



Handle HTTP Status 404 & 403

Untuk menangani error "Server Not Found" atau lebih tepatnya **404 Not Found** dan **403 Forbidden** di server Ubuntu 22.04 menggunakan .htaccess, Kita dapat mengikuti Langkah-Langkah dibawah ini:

[1] Aktifkan Modul Rewrite Apache:

- `sudo a2enmod rewrite`
- `sudo systemctl restart apache2`

[2] Pastikan .htaccess Diizinkan:

- `sudo nano /etc/apache2/sites-available/000-default.conf`

Tambahkan bagian ini di paling bawah konfigurasi:

```
<Directory /var/www/html>  
    AllowOverride All  
</Directory>
```

[3] Buat atau Edit File .htaccess:

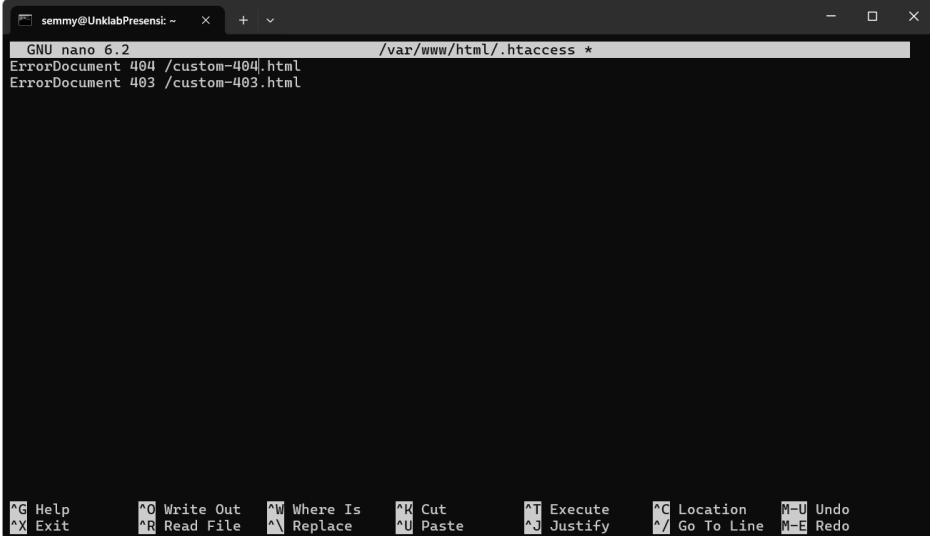
- `sudo nano /var/www/html/.htaccess`

Add kode berikut untuk menangani error 403 dan 404:

EditErrorDocument 403 /custom-403.html

EditErrorDocument 404 /custom-404.html

- **Ctrl+s** untuk save file
- **Ctrl+x** untuk keluar dari nano text editor.
- `sudo systemctl restart apache2`



Mengarahkan semua halaman yang tidak ditemukan ke file custom-403/404.html.

Create Custom File 403/404

Buka putty dan run script di bawah ini:

- `sudo su`
- `sudo nano /var/www/html/custom-403.html`
- `sudo nano /var/www/html/custom-404.html`
- **Ctrl+s** untuk save file
- **Ctrl+x** untuk keluar dari nano text editor.

Membuat file
«**custom-403.html**»:

```
root@UnklabPresensi: /home. x + v
GNU nano 6.2 /var/www/html/custom-403.html *
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>403 - Forbidden</title>
</head>
<body>
  <h1>403 - Forbidden</h1>
  <p>You do not have permission to access this page.</p>
  <a href="/">Back to Home</a>
</body>
</html>
```

GANTI BUTTON:
`<a onclick="window.history.back();" class="home-button">Back to Home</a>`

```
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

Membuat file
«**custom-404.html**»:

```
root@UnklabPresensi: /home. x + v
GNU nano 6.2 /var/www/html/custom-404.html *
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>404 - Page Not Found</title>
</head>
<body>
  <h1>404 - Page Not Found</h1>
  <p>The page you are looking for is not available.</p>
  <a href="/">Back to Home</a>
</body>
</html>
```

GANTI BUTTON:
`<a onclick="window.history.back();" class="home-button">Back to Home</a>`

```
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

Error Handling

Jika .htaccess tidak berfungsi, periksa apakah file tersebut memiliki izin yang sesuai:

➤ `sudo chmod 644 /var/www/html/.htaccess`

Jika tetap tidak berhasil, cek error log Apache:

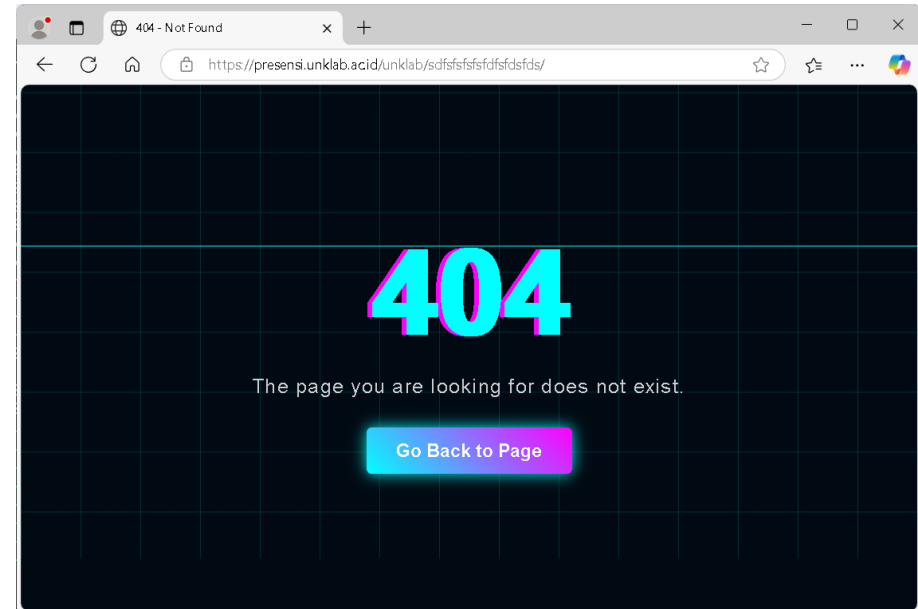
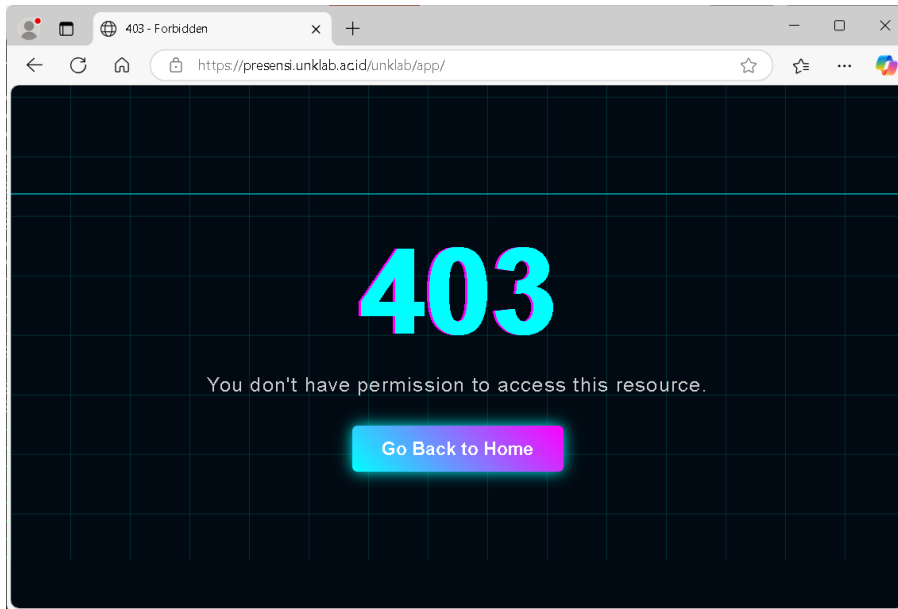
➤ `sudo tail -f /var/log/apache2/error.log`

Expected Output

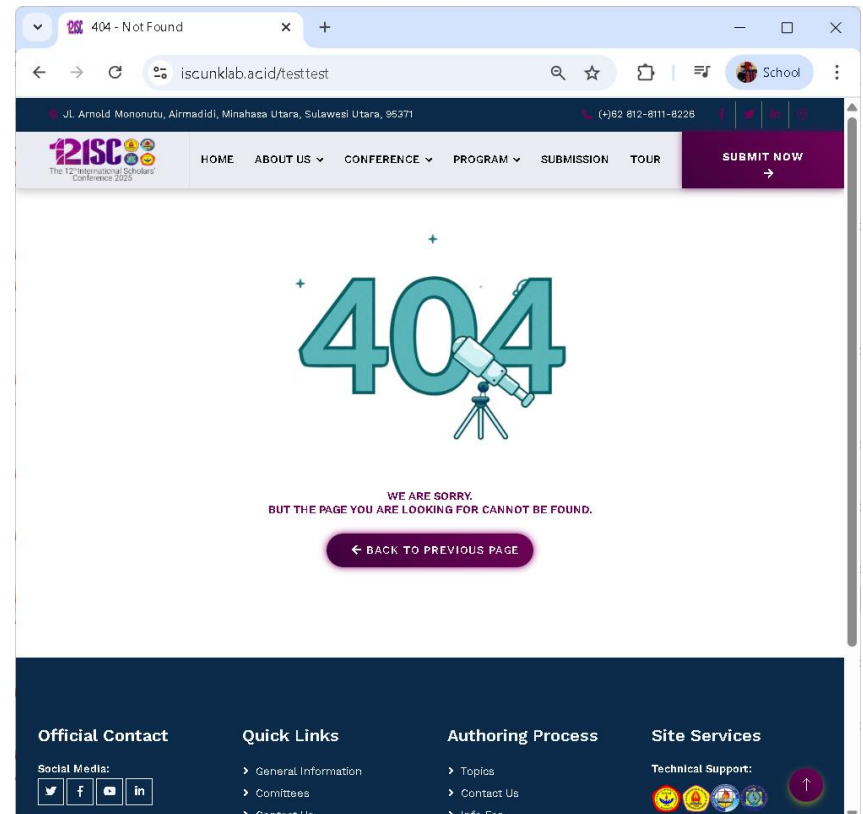
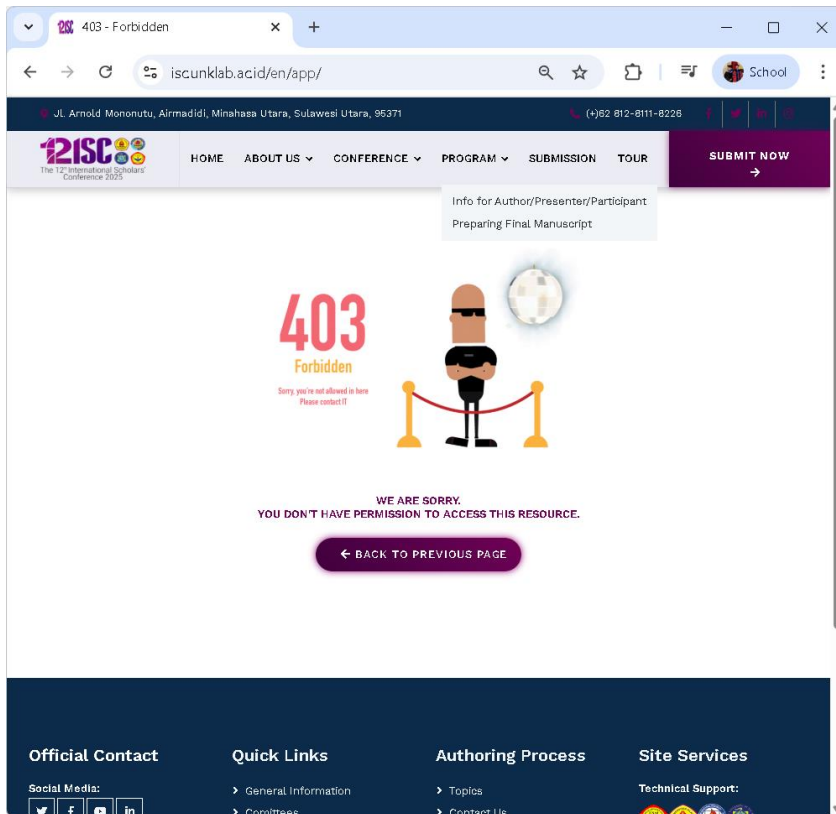
Coba akses halaman yang tidak ada di website,

Contoh 403 (Forbidden) : <http://yourdomain.site//unklab/app/>

Contoh 404 (Not Found) : <http://yourdomain.site/sdfsfsfsfsfsfsf>



Buatlah versi Custom File 403/404
kalian sendiri tapi harus memiliki dashboard,
header dan footer.



END PRESENTATION

Thank you for your attention

Instructor: S – W – T

THANK YOU

